

DIVERSITY AMONG PLANTS

Chapter

8

Major Concept

In this Unit you will learn:

- Evolutionary Relations in Plants
- Non-Vascular Plants
- Seedless Vascular Plants
- Seed Plants



Diversity among Plants

Plants are multicellular eukaryotes with Photosynthetic nutrition. Cell typically have cellulose wall, sap vacuole, plastids and several photosynthetic pigments which always include chlorophyll-a. They have tendency to develop embryo and have heteromorphic alternation of generation. Sexual and asexual both types of reproduction are reported in plants.

8.1 THE EVOLUTIONARY RELATION IN PLANTS:

Although life probably began on earth about 3.5 billion years ago. The earliest plants have become the first organisms adapted for life on land. In the beginning the plants were restricted only to aquatic habitat but gradually they adapted to terrestrial habitat.

The evolutionary history of the plants kingdom is a story of adaptation to changing terrestrial condition.

There are wide variety of plants species exists on earth. They live in every type of habitat .They have colonized great areas of the earth surface in practically all sorts of soil and climatic conditions. They show enormous diversity in size and form.

Cuticular covering of the plant body, protective layer around the sex organs, evolution of an internal transport system of xylem and Phloem, specialised organs, such as leaves and roots and woody tissue allowed vascular plants to attain greater body size than smaller non-vascular plants, which rely on simple diffusion for transport of water and nutrients are the important characteristics in the transition of plant from water to land. Transport of male gametes in pollen to the female reproductive organ. The pollen grain dispersed by wind or animals rather than by water.

Although the earliest vascular plants were relatively small and probably, confined to swampy areas, after that a rapid diversification of plants occurred gradually.

"Plants may be defined as multicellular eukaryotes that are photosynthetic autotrophs (chlorophyllous) with cell wall primarily made up of cellulose , exhibiting heteromorphic alternation of generation and zygote retained to become develops into embryo".

8.1.1 General Characterstics of Plants:

- Plants are multicellular eukaryotes.
- They are Photosynthetic autotrophs and contain photosynthetic pigments specially chlorophyll, which enable the plants to convert light energy into food (chemical energy).
- Rigid cell wall of plant cell primarily made up of cellulose
- Life cycle of show heteromorphic alternation of generation.
- Zygote retained and develops into embryo



- Plant cuticle, composed of insoluble lipid polymers and waxes, barriers to prevent diffusion of water to the drier atmosphere and also filters a substantial component of UV radiation from sunlight.

Some other characteristics included are:

Plants produce some secondary products like sporopollenin and lignin. Sporopollenin is tougher than lignin, composed chiefly of carotenoids. It protects them from attack by grazing organisms and microorganisms. It also makes spores of land plants tough and flexible and resistant to biodegradation. Lignin is important to provide strength to upright terrestrial plants.

In terrestrial environments, gases enter through small pores in the plant surface. Stomata are present in modern hornworts, mosses, and vascular plants as well as simple, ancient fossil plants such as *Rhynia*.

8.1.2 Vascular supply and lignin-Transport support:

The vascular transport of plants consists of xylem and phloem for the transport of water and mineral ions and for the transport of sugar respectively. Another product, lignin, provides strength to upright terrestrial tissues.

8.1.3 Alternation of generation and sexual reproduction:

Alternation of generation is the type of life cycle where asexually reproducing generation alternates with sexually reproducing generation. It is an adaptation. There are two alternating generations in the plant life cycle. The gametophyte generation, which is haploid, produces male and female gametes by mitosis. Male and female gametes fuse to form a zygote, which grows into the diploid sporophyte generation. The diploid sporophyte produces spores by meiosis, which are dispersed and germinate into new haploid gametophytes. In land plants, haploid and diploid generations are markedly heteromorphic; these gametophyte and sporophyte differ in size and shape.

8.2 NON-VASCULAR AND VASCULAR PLANTS

Non-Vascular Plants:

The non-vascular plants are the group of plants without specialized conducting tissues (xylem and phloem) for transporting water and nutrients. The gametophyte is the dominant stage of life cycle in these plants.

They need moist habitat to reproduce because flagellated sperm must swim through water drops, e.g. Liverwort, hornwort, and mosses.

8.2.1 Bryophytes:

Bryophytes are the simplest non-vascular land plants, first appearing before 450 million years ago. In bryophytes, strengthening and conducting tissue is absent or poorly developed. It includes Liverworts (Class Hepaticae), Hornworts (Class Anthocerotae), and mosses (Class Musci).

Bryophytes are small photosynthetic free-living gametophytes that lack vascular tissue. Sexual reproduction requires water for sperm to swim from an antheridium to an egg in an archegonium. The sporophyte grows on the gametophyte and is dependent on it for nutrition.

General Characteristics:

These are non-vascular plants, where xylem and phloem are absent.

- Plants showing alternation of generation with dominant gametophytes having amphibious nature
- Gametophytes are the conspicuous, chlorophyllous, photosynthetic autotrophs having thalloid body or differentiated in rhizoids, Pseudo-stem and pseudo leaves.
- Male gametes (Antherozoid or spermatozoids) are produced in antheridium (male gametangia, female gametangia). while female gametes (non-motile egg cell) is produced in archegonium.
- Sporophytes are semi-parasite on gametophytes having a body differentiated into foot, seta and capsule. Spores are produced by the sporophytes from spore mother cell in capsule by meiosis.
- They live mainly in damp, shady places.
- Bryophytes are amphibious in nature, need water to reproduce. Their sperms are flagellated and swim from antheridium to archegonium to fertilize the egg when archegonia mature and produce smell. The male gametes mature earlier than female gametes this property is called protoandry.
- Bryophytes lack the lignin fortified tissues required to support tall plants on land.

8.2.2 Life cycle of moss:

Life cycle of Bryophyte (Moss)

All bryophytes show heteromorphic alternation of generation with gametophytes as dominant generation including *Funaria hygrometrica* (a moss). The gametophyte is haploid consisting of rhizoids, pseudostem and pseudo leaves.

The sex organs called **antheridia** (male) and **archegonia** (female) develop at the tips of pseudo stem which are always **dioecious** having either male or female sex organs. There is **protoandry** because antheridia mature earlier and liberate their antherozoids, which start swimming with the help of their flagella in dew or rainwater. When archegonia mature, they have single ovum in the **venter** and few neck canal cells in the neck. Swimming sperms are attracted by scent of sugar cane secreted by mature archegonium but a single antherozoid fuses with the ovum to form **diploid** (2n) **oospore** (zygote). This is retained within archegonium and form an **embryo** (2n). This embryo undergo repeated mitotic-divisions to form sporogonium (a sporophyte) which is diploid. It consists of foot, seta and

capsule. Within capsule spore mother cells are present. Each spore mother cell divides by meiosis to form four haploid spores. Each spore when drop at proper substratum germinates into a filamentous body called **protonema**, later on gametophyte (haploid) develops from protonema to complete life cycle.

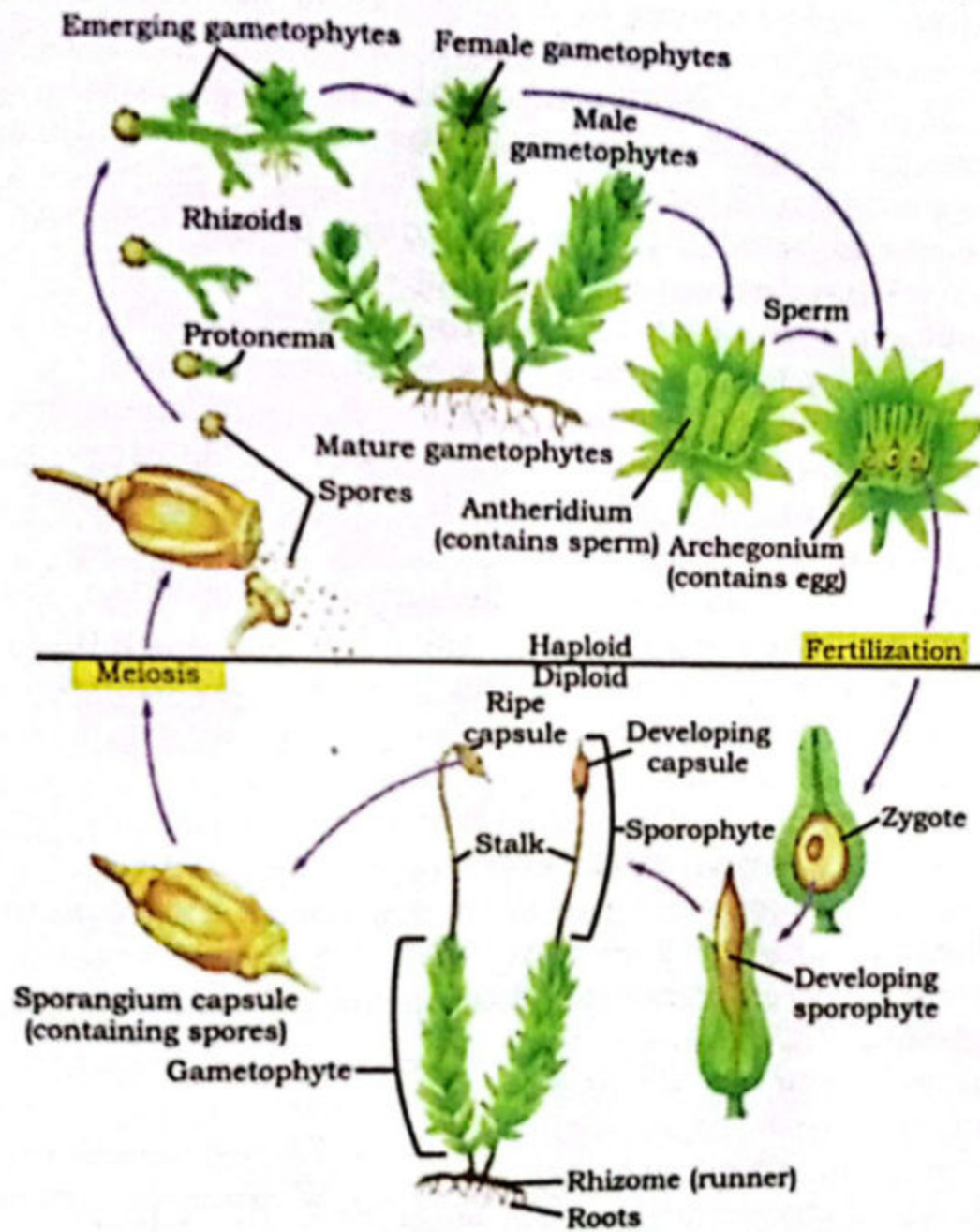


Fig 8.1: Life cycle of Moss

8.2.3 Adaptations to land habitat:

The first evidence that plants had invaded the land from the sea is found in fossils. All the biologists agree that the land plants and animals evolved from aquatic ancestors. The conquest of the land must have been a long and difficult process. The plants had to become adapted by developing new structures.

Life of aquatic organisms is an easy life. Water is necessary for the growth of all living things and there is little danger in the sea of any lack of

water. Carbon containing compounds, so essential for autotrophs are present abundantly in solution. These autotrophs in turn provide a continuous supply of oxygen for all the living organisms in the sea. The temperature in the seas does not fluctuate as much as temperature on land. Hence, the aquatic environment is more uniform and better supplied with some of the necessities of life than in the rigorous land environment.

When plants invaded the land from the sea. They faced many problem such as obtaining and conserving water, absorbing carbon dioxide from the atmosphere for photosynthesis.

To solve these problems, land invading plants adopted themselves first to amphibious-habitat and later developed a complete terrestrial form of life. The amphibious form of land plants includes all the bryophytes. We will consider the following adaptive characters exhibited by them.

- | | |
|-------------------------------------|--------------------------|
| 1. Rhizoids for water absorption | 2. Conservation of water |
| 3. Absorption of CO_2 | 4. Heterogamy |
| 5. Protection of reproductive cells | 6. Formation of embryos |

1. Rhizoids for water absorption:

The study of *Marchantia* thallus and other bryophytes show that they have rhizoids for water absorption. These are long, filamentous extensions of the cell of the lower surface of the thallus. They greatly increase the surface for absorption of water from the soil.

2. Conservation of water:

To conserve water within the body thallus of all bryophytes is multilayered and very small surface area directly exposed to reduce the drying effects of the atmosphere. Moreover, the outer and uppermost layer of cells is covered with cuticle. It is non-cellular layer of wax-like substance called cutin. This is very efficient in reducing the rate of evaporation.

3. Absorption of CO_2 :

Land plants need an efficient means for the exchange of gases with the environment in contrast to aquatic plants which exchange gases dissolved in water. The upper surface of the thallus is provided with a number of aerating pores. Each pore leads inside into an air-chamber. These chambers have filamentous branching photosynthetic cells. Air having CO_2 enters through the pores and CO_2 from air absorb by these photosynthetic cells and diffuses into the cytoplasm.

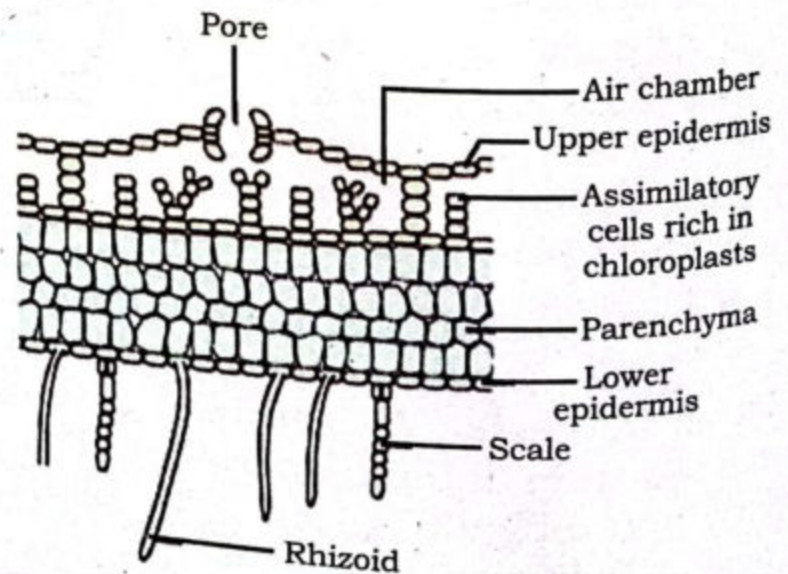


Fig 8.2 *Marchantia* thallus



Heterogamy:

4. Heterogamy is the most successful kind of reproduction that has evolved in bryophytes. It is defined as production of two different types of gametes. One is male (motile) and the other is female (non-motile) full of stored food.

Protection of reproductive cells:

5. The land environment requires special protection for the reproductive cells. In amphibious plants reproductive cells are very well protected. The male gamete (sperm) and female gamete (ovum) are produced in multicellular reproductive sex organs called **antheridia** and **archegonia** respectively. Moreover, these organs are also surrounded by hair like structures called **paraphyses** which help to prevent drying of the sex organs.

Formation of embryos:

6. Embryo formation in amphibious plant is of universal occurrence. The fertilized egg called oospore (zygote) is formed inside the archegonium. An embryo develops from the oospore as it divides still inside the protective coverings of the archegonia. Thus the coverings formed by the female organ protect the growing embryo from drying out and from mechanical injury.

8.2.4 Advantages of Bryophytes:

Bryophytes play important role in maintaining the ecosystem. They initiate soil formation, maintain soil moisture, and in recycling of nutrients in forest ecosystem. They are bio indicators of heavy metals in air pollution.

From ancient time in many parts of the world bryophytes used as medicines for example Liverworts (*Marchantia*) used for the treatment of liver disorder, another member of bryophytes *Polytrichum* used for curing diseases such as fever, homeostatic and traumatic injuries to pneumonia and lymphocytic leukemia. Insecticidal activities of bryophytes are also reported.

8.3 VASCULAR PLANTS:

Vascular plants characterized by the presence of vascular tissue. Vascular tissue is made up of Xylem and phloem. Vascular tissue has two important properties. First these tissues are concerned with translocation of water and nutrients throughout the plant body. Xylem carries mainly water and mineral salts, whereas Phloem carries mainly organic solutes in solution, such as sugar. Secondly plant bodies can be supported because Xylem contains lignified cells of great strength and rigidity. Another lignified tissue, sclerenchyma also develops in vascular plants and supplements the mechanical role of xylem. The dominant generation of living vascular plants is the sporophyte existing as free-living plant. The gametophyte is smaller in size, bear smaller sex organs. It may be monoecious (Bisexual) or dioecious (Unisexual), short lived, and degenerates once the sporophyte is established.

Vascular plants are categorized in two major groups.

- Seedless plants---- Fern allies (Psilopsida, Lycopsidea and Sphenopsida) and ferns.
- Seeded plants (Spermatophyte) ---- Gymnosperms and Angiosperms.

8.3.1 Seedless Vascular Plant(Pteridophytes):

Seedless vascular plant include fern allies and ferns.

Fern allies:

The term " Fern allies" refers to a loose assemblage of spore producing vascular plants that are classified in three group.

- Sub division Psilopsida e.g Psilotum and tmesipteris.
- Sub division Lycopsidea e.g Lycopodium and Selaginella
- Sub division Sphenopsida e.g Equisetum

(i) Sub division Psilopsida (Whisk ferns) living fossils:

This is the most primitive group of living vascular plants includes two genera Psilotum, Tmesipteris. The Psilopsida was considered as living fossil because of its apparent similarities to the fossil plant Rhynia. Psilotum is the only living vascular plant where leaves and roots are absent.

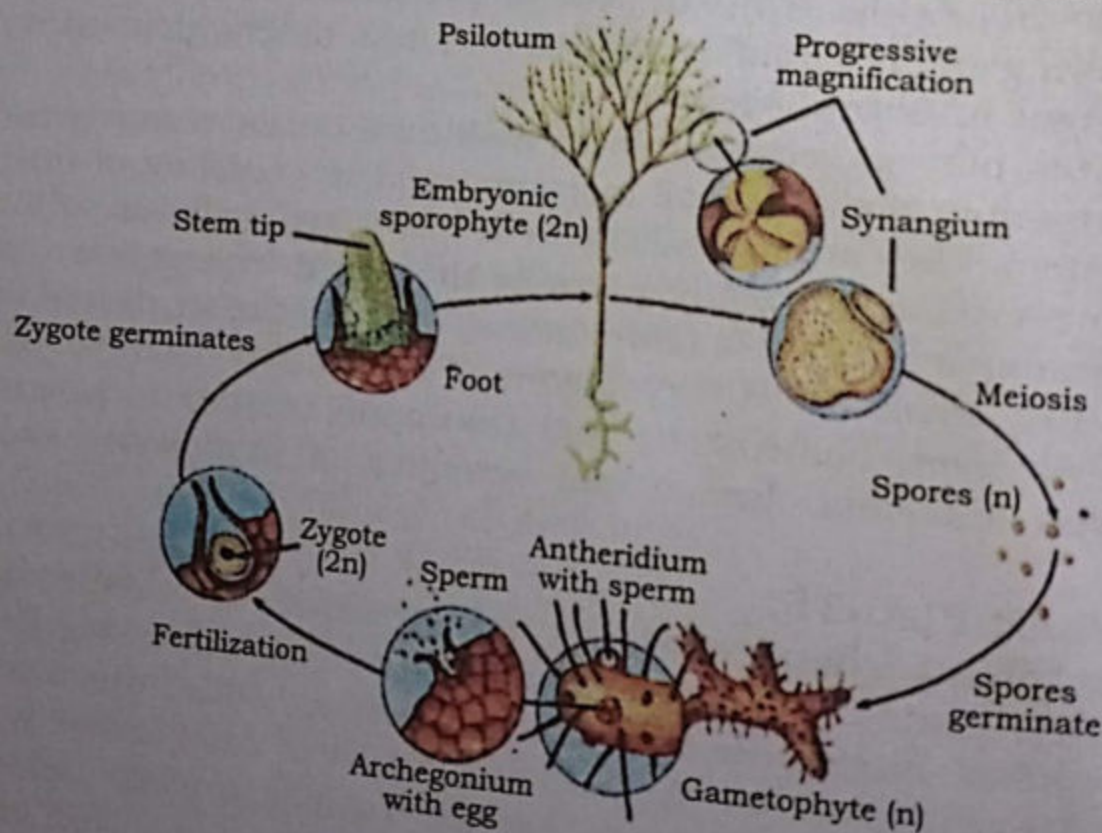


Fig 8.3 Life cycle of psilotum

The Psilopsida sporophytes are simple dichotomously branching small Plants. They lack true leaves but produce small out growths of stem (enations). The aerial stems are green and carry out photosynthesis. Sporangia develop at the tips of some of the aerial branches. Within the sporangia meiosis produces haploid spores from spore mother cells. These

spores are morphologically similar therefore called homosporous. Each spore germinates into gametophytes of Psilopsida which is also called prothallus. It has one or more dichotomous branches. It bears numerous unicellular rhizoids. It is monocious. The male sex organ antheridium and female sex organ archegonium are produced near the growing apex.

(ii) Sub division Lycopsidea:

They are commonly called as club mosses most of which are tropical. They are the first plants to have evolved true roots. It is generally supposed that these roots arose from branches of the ancestral of algae that penetrated the soil and branched underground. The sporophyte consists of dichotomously branched shoots that bear simple spike like leaves. Sporangia are located in leaf axil either along normal portions of stem or condensed into cones, strobilus. These sporangia containing leaves are called sporophyll. The lycopodium are homosporous having one type of spores develop monoecious gametophytes that bears both male and female sex organs. *Selaginella kraussiana* is a common green house plant. *Selaginella* exhibits heterospory i.e., it produces spores of two distinct kinds large haploid spores germinate to produce female gametophytes, which archegonia development on it and small microspores which develop into male gametophytes, antheridium on it. These gametophytes are dioecious.

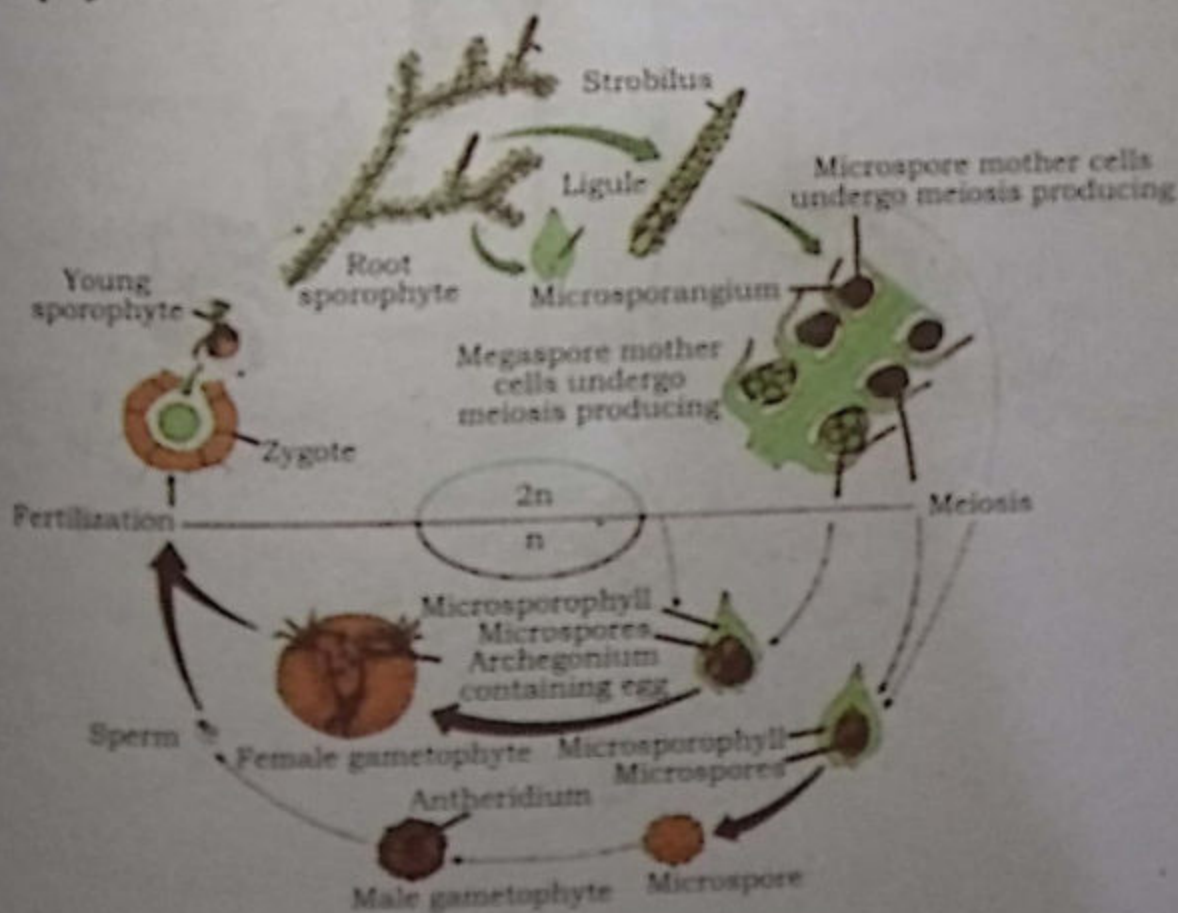


Fig 8.4 Life cycle of *Selaginella*

(iii) Sub division Sphenopsida (Horse tails):

Horsetail grow abundantly in the swamps of carboniferous times. Today the phylum consists of only one living genera *Equisetum*. It consists of a creeping underground rhizome that produces upright stems. Stems are green and photosynthetic. Small leaves are borne in sheath like whorls at nodes in the stem. Sporangia are produced in small strobili borne terminally on either normal vegetative or specialized reproductive shoots. Following meiosis, spores are produced. Spores are unusual. The outer layer of spore wall form four sporopollenin thread called elators their movement assists in spore dispersal. When a spore lands in a damp place, the elator coil up and spore germinates. A tiny gametophyte about the size of a few millimeters up to 3 centimeter is produced.

Gametophyte is dorsiventral. Unicellular rhizoid are arised from the basal cells. Gametophyte is basically bisexual. It bears both male sex organ (antheridia) and female sex organ (archegonia).

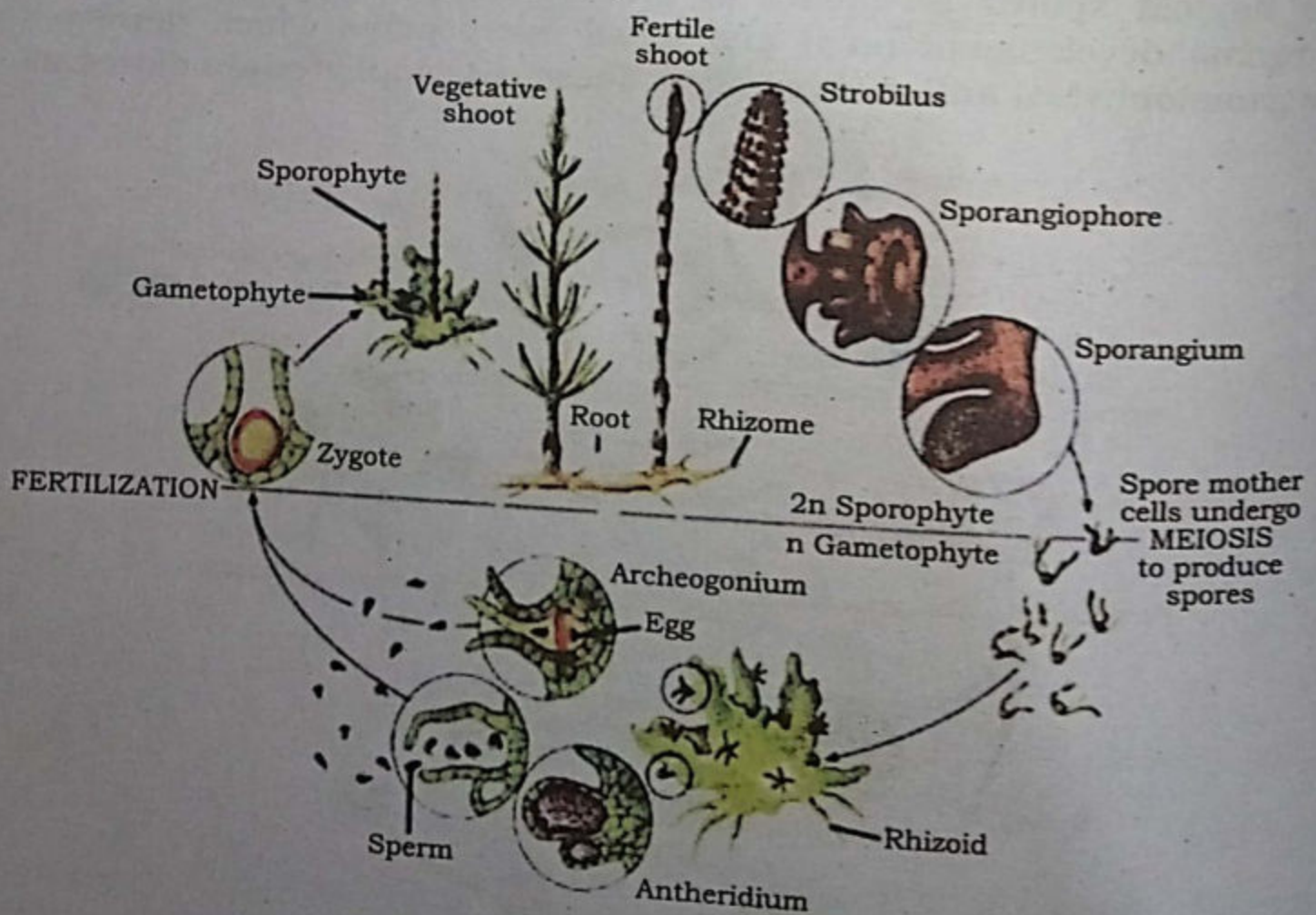


Fig 8.5 Life cycle of *Equisetum*



(iv) Sub division Filicinophyta (Ferns):

Ferns are the most diverse and conspicuous of the spore – producing vascular plants. Most of these are found in tropical regions but many species grow abundantly in damp habitat in temperate regions. They range in size from small, delicate filmy ferns and floating aquatic to tall trees.

Ferns shows heteromorphic alternation of generation in which the sporophyte is dominant. The sporophyte generation possesses true roots, stems and leaves. Vascular tissue (Xylem and Phloem) present in the sporophyte. Leaves relatively large and called fronds. Spores produced in sporangia which are usually in clusters called sorus.

Gametophyte is reduced to a small, simple prothallus. The prothallus contains male reproductive organ antheridia and female reproductive organ archegonia. e.g. *Dryopteris*.

8.3.3 Evolution of the leaf:

The leaf is the most important organ of a green plant because of its photosynthetic activity. There are two types of leaves in vascular plants.

- (a) One veined leaf (microphyllous)
- (b) Many veined leaf (megaphyllous)

It is very interesting to trace the origin of leaf in the green plants.

Evolution of One Veined Leaf:

Enation Theory:

The evolution of one-veined leaf (microphyllous) can be explained by assuming that a thorn like outgrowth (Enation) emerged on the surface of the naked stem. With an increase in size of the out growth, the vascular tissues were also formed for the supply of water, food and support to the leaf.

Reduction Theory:

Another possibility is that a single veined leaf originated by a reduction in size of the part of the leafless branching system of the primitive vascular plant. This is how the leaf of lycopodium (club mosses) and equisetum (horse tail) came into existence.

Evolution of Many Veined Leaf:

Webed Theory:

Many veined leaf (megaphyllous) originated much later. These are the evolutionary modifications of the forked branching system in the primitive plant. The first step in the evolution of this leaf was the restriction of forked

branches to a single plane. The branching system became flat. The next step in the evolution was filling the space between the branching and the vascular tissues. The leaf so formed looked like the web foot of a duck

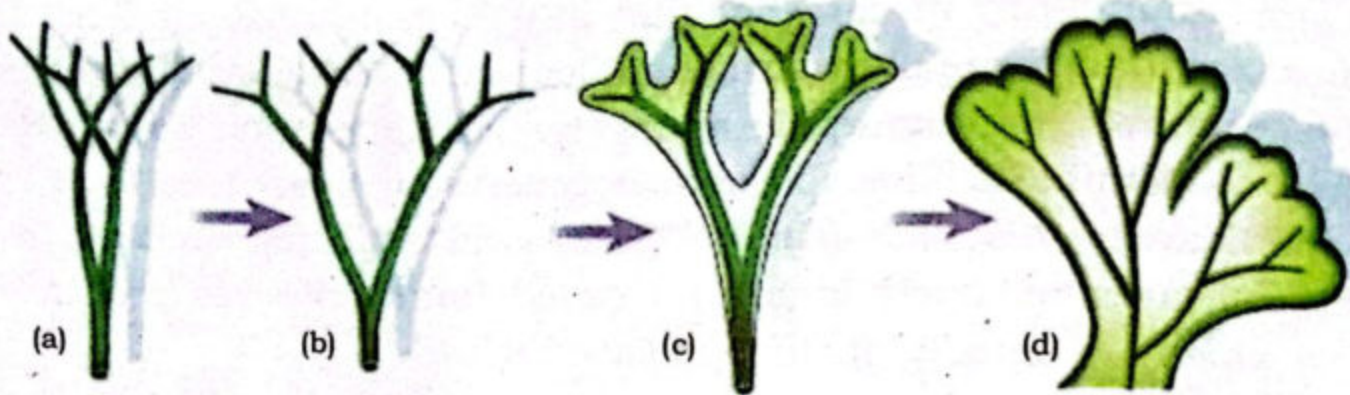


Fig: 8.6 Evolution of leaf

8.3.3 Life cycle of a fern:

The life cycle of fern (*Adiantum*) or *Dryopteris* shows heteromorphic alternation of generation in which sporophytic phase is dominant. All ferns are homosporous, producing single types of spores.

Sporophytic Phase:

The sporophyte ($2n$) which is diploid, consists of adventitious roots, underground stem a rhizome and pinnately compound leaves.

During asexual reproduction the leaves give rise to rounded bodies on its undersides which are called sori (singular sorus). The sori are green but when ripe they become dark brown. The leaves which bears sori are called sporophylls. Each sorus is the group of sporangia.

Sporangium consists of a stalk called **sporangiophore** and a biconvex capsule consisting of **annulus** and **stomium**. The annular cells are thickened whereas stomial cells are thin-walled. Within sporangium spore mother cells are present. Each spore mother cell divides by meiosis to form four haploid spores. The spores are liberated through stomium.

Each spore on germination gives rise to miniature bisexual (monoecious) gametophyte called **prothallus**.

Gametophytic Phase:

The gametophyte of fern is called prothallus. The prothallus is independent autotrophic, heart shaped, dorsoventrally flattened lying prostrate on same wet substratum. It is fixed soil with the help of rhizoids which absorb water and nutrients. The prothallus is monoecious having archegonia and antheridia on the same prothallus.

Archegonium consists of venter with an ovum and a neck and secrete malic acid by degenerating neck canal cells at maturity. Each antheridium produces a number of antherozoid (sperms). A number of sperms, by making chemotactic movement in water reach to the archegonium. Only one



sperm fuses with ovum to form oospore (zygote) which is diploid. Young sporophyte develops from the oospore. In the meantime prothallus degenerate in this way life cycle is completed.

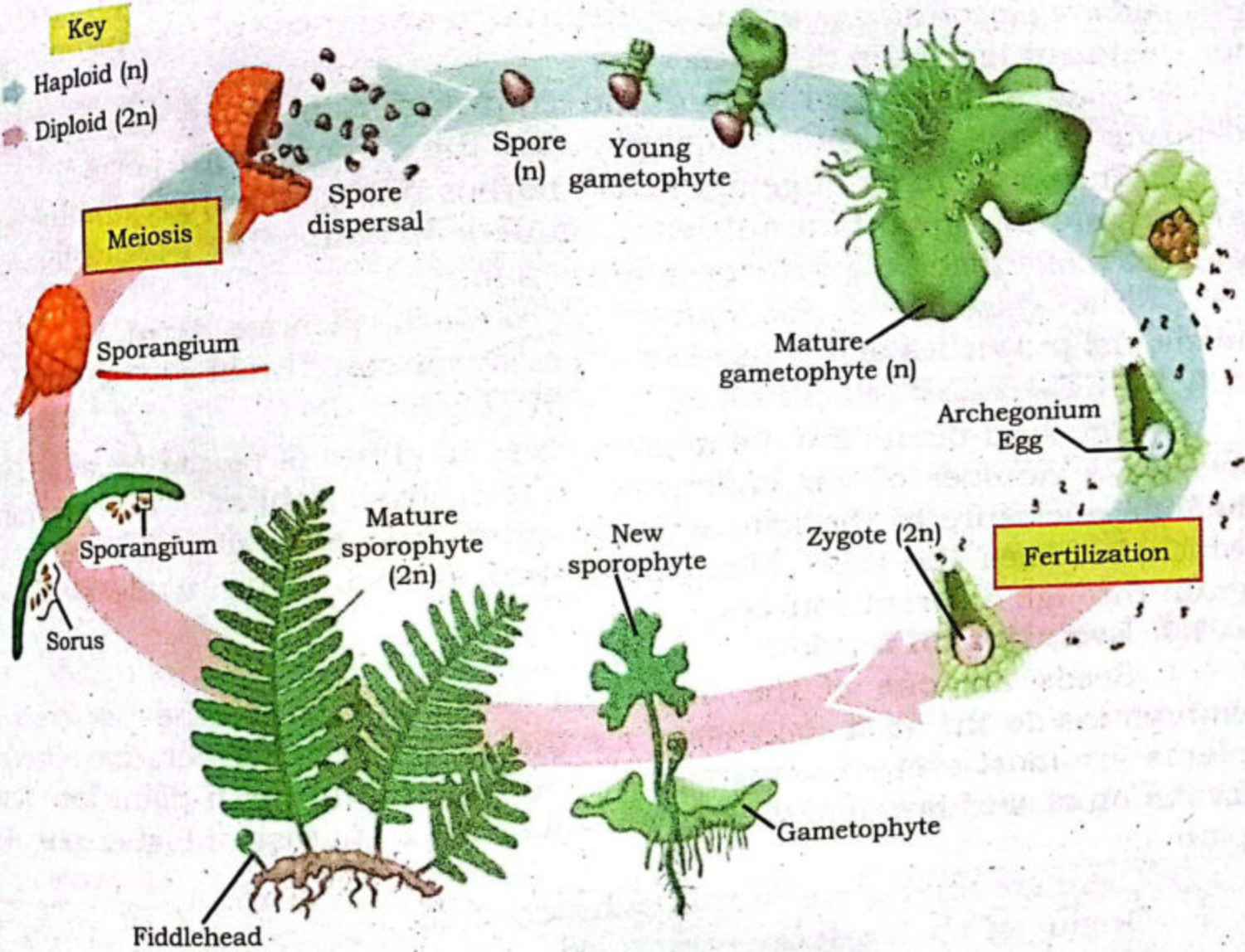


Fig: 8.7 Life cycle of fern

8.3.4 Vascular Plants as successful land Plants:

The vascular plants have evolved a number of adaptations to the terrestrial environment that have enabled them to invade all the most inhospitable land habitat. During this process they diverged sufficiently from one another. The four fundamental adaptations of vascular plants (with a few minor exceptions) make them more successful land plants.

- A protective layer of sterile jacket cells around the gametangia.
- Multicellular embryo retained within the archegonia.
- Cuticles on the aerial parts to prevent evaporation of water.
- Xylem for conduction of water and, dissolved minerals as well as provide support.

Many other such adaptations, absent in the earliest vascular plants, appear in most advanced member of the division.

8.3.5 Importance of Seedless Vascular Plants:

Seedless plants have historically played a role in human life. Man get many advantages including food, shelter, fuel and medicine and decoration. They had been used as food e.g. *Pteris*, *Ceratopteris* and *Marsilea*. *Polypodium glycorrhiza* is a part of diet due to sweetness of its rhizome, and for treatment to certain diseases.

Coal, a major fuel source and contributor to global warming, was deposited by the seedless vascular plants of the carboniferous period.

The water ferns of genus *Azolla* harbor nitrogen fixing cyanobacteria and restore this important nutrient to aquatic habitats. The beautiful fronds of ferns make them a favorite decorative plants.

The rhizome of *Polypodium glycorrhiza* (licorice fern) also have medicinal properties and is used as a remedy for sore throat.

8.4 SEED PLANTS: (SPERMATOPHYTES)

The seed plants are the most successful group of plants because they evolved a number of adaptations to a terrestrial habitat. These features include enclosure of the female gametophyte and embryo within an ovule which becomes the seed. Microspores transported to the ovule as pollen grain through different sources.

8.4.1 Evolution Of Seed:

Seeds are one of the important parts of plants, the presence of embryo inside the seed, necessary for the continuity of generation. Seeded plants are most evolved, advanced and successful group of plant kingdom. Evolution of seed is one of the key reasons for the success of higher vascular plants.

There are three steps in the evolution of seed.

- Origin of heterospory i.e., two types of spores are produced. Microspores grows into sperm forming gametophyte (male gametophyte) and megaspore grows into egg forming gametophyte (female gametophyte).
- Development of integument, an enveloping structure for the protection of megasporangia to form ovule.
- Retention of the mature megaspores in the sporangia to develop female gametophyte

The development of seed has given the vascular plants better adaptation to their environment.

8.4.2 Groups of Seeded Plants: (Spermatophytes)

The seed plants have traditionally been divided into two groups

1. The Gymnospermae
2. The Angiospermae



The Gymnospermae:

They are the successful group of plants of worldwide distribution, accounting for about one third of the world's Forest

General Characteristics of Gymnosperms:

- They have naked seeds because ovules are not covered by ovary i.e. ovary absent.
- They are trees or shrubs mostly ever green with needle like leaves with xerophytic adaptations.
- They are cold resistant and found in colder parts of the world.
- The tree is the sporophyte generation heterosporous producing two types of spores. The best-known group of gymnosperm is the conifers. In conifers ovules (later seeds) are located on the surface of specialized scale leaves called ovuleiferous scales. These are arranged in cones. Male and female cones are produced on the same tree.
- Most conifers bear their micro and megasporangiate structure in cones.
- Gametophyte generation reduced, the functional megaspore develops to form miniature female gametophyte which is permanently lodged within megasporangium (Ovule)
- Microsporangia (Containing pollen) are borne on microphylls (Fertile leaves).
- Vascular tissue containing Xylem tracheids and Phloem sieve cells while Xylem vessels and phloem companion cells are absent.
- Female gametophyte consists of two to five archegonia each having single ovum.

Coast red grow to more than 100m. The trunk of the general sherman tree is estimated to weight 625000 kg.

The largest trees in the world, *Sequoia gigantea* of south western united states belonging to the gymnosperms.

They are cold resistant and found in colder regions.

- Each microspore develops into another miniature male gametophyte consisting of stalk nucleus tube nucleus, two male gametes and two prothelial cells within an elongated pollen tube.
- Pollen grains of conifers are wind dispersed.
- Fertilization takes place within sporophyte.
- After fertilization, megasporangium gives rise to seed.
- Seed undergoes epigeal germination to form new sporophyte plant. e.g. Pinus, the familiar tree, fir, larch, cedar redwood, Yew.

Conifers are the most diverse and wide spread of the living non flowering seed plants. They are all woody plants many of which are important soft wood timber tree.

Uses of Gymnosperms:

- Conifers are commercially important, conifer wood is valuable commercial resource world wide. Pines, spruce firs and other conifers are a major source of softwood timber used in house construction, the timber can be pulped to make paper or compressed to make chipboard, the basic material for modern furniture.
- In addition, Pines are a source of resins, turpentine and pine oil.
- The plants are widely used as ornamentals and largely spotted in gardens and parks e.g Cycas, Thuja, Juniperus, Ginkgo.
- The famous dry fruits locally called chilghoza (*Pinus gerardiana*) belongs to pine tree.
- Ephedrine a drug obtained from Ephedra is used in various respiratory disorder particularly in asthma.

8.4.3 The Angiosperms:

The flowering Plants, or angiosperms have their seeds enclosed in fruit because ovules are covered by ovary. They are dominant vascular plants of modern flora, adapted to a great range of habitats from the tropics to polar regions, including fresh and salt water.

The smallest plants are free floating aquatic duck weeds which are highly reduced and not even differentiated into leaf and stem. The tallest flowering plant is the mountain ash tree.

One of the most important features of angiosperm is a part from the enclosed seed is the presence of flowers as reproductive structure instead of cones.

Flowers are unique to angiosperms and have become adapted to attract insects or other pollinators, for example by the development of large brightly coloured petals, nectaries and scent glands.

Flowering plants are classified into two major groups the monocotyledonae and the dicotyledonae.

Difference between Monocotyledonae and Dicotyledonae:

Monocotyledonae	Dicotyledonae
Seeds Plants produce only one cotyledon (Scutellum) containing seed.	Plants produce two cotyledon (seed leaves) containing seed.
Leaf Veins are parallel. Dorsal and ventral surfaces are same. (Monofacial leaf)	Net like pattern of veins (Reticulate Venation). Dorsal and ventral surfaces are different (bifacial leaf)
Stem Stem is mostly unbranched, vascular bundles scattered, vascular cambium. Cambium usually absent so no secondary growth to (exceptions occur e.g palms)	Stem is mostly branched, vascular bundles arranged in a ring. ring usually present giving rise secondary growth.



Root
Adventitious roots from the base of the stem take over from the primary root, giving rise to a fibrous root system

Roots are mostly branched tap roots.

Flower
There are three whorls in flowers. No distinct petals and sepals. These structures are combined to form "Perianth Segment" e.g Grasses, Iris, Onion, Maize, Palms

There are mainly four or five whorls. Usually distinct petals and sepals e.g Pea, Rose Plant, Daisy, Acacias, Nerium.

8.4.4 Life cycle of flowering Plants:

Flowering plants show heteromorphic alternation of generation. The haploid gametophyte alternates with diploid sporophyte. The gametophyte generation is reduced the main plant body is sporophyte.

In the life cycle of plant, meiosis produces female and male haploid spores, each of which undergoes cell division by mitosis to produce a multicellular haploid stage, the female or male gametophyte. These gametophyte produces haploid gametes by mitotic division. Two gametes fuse to form a diploid zygote which then undergoes mitotic divisions to produce the diploid life stage, the sporophyte.

Sporophyte generation:

Asexual reproduction in Angiosperm:

The main plant body is sporophyte that consist of root, Stem, leaves and flowers. Root, stem and leaves are considered as vegetative parts while flower is the reproductive part. Flower is a compressed reproductive shoot with four whorls of modified (floral) leaves called sepals, petals microsporophyll (stamen) and megasporophyll (carpel). Microsporophyll (stamen) is male reproductive structure and megasporophyll (carpel) is female reproductive structure.

Microsporophylls (Stamens):

Microspores which later develops into pollen grains are produced by **microsporophyll** or **stamens**, each stamen consist of sac like anther and a stalk called filament, usually the anther have four chambers called **pollen sac** or **microsporangia**, contain numerous microspores mother cells.

Each microspore mother cell produces four microspores by meiosis so each pollen sac has numerous microspores, develop into pollen grains and disperse from here for pollination.

Megasporophylls (Carpels):

Megasporophyll now called carpel produce ovum in ovary. Each carpel consist of three parts, stigma, style and ovary, within ovary one or more ovules (megasporangium) develop. The main body of ovule is called

nucellus which is protected by two layers of protective sheaths of cell called **integuments**. A small pore is present on it, the **micropyle**, with these it contains one megaspore mother cell which produce four megaspores on maturation. Out of four megaspores only one megaspore will be functional to develop into **female gametophyte**. The female gametophyte develops inside ovule called **embryo sac**.

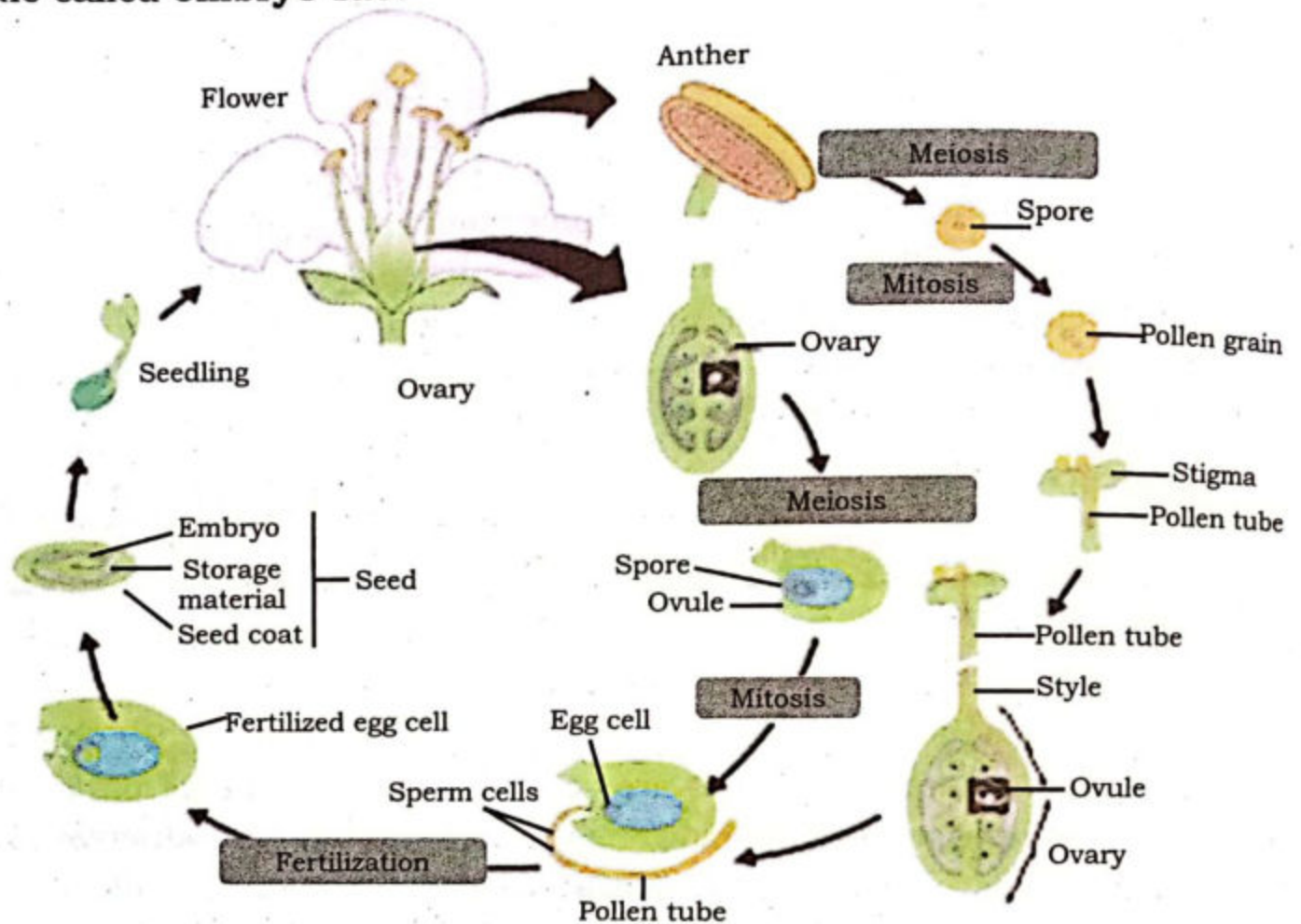


Fig 8.8 Life cycle of Angiosperms

Development of female gametophyte:

Megasporangium (Ovule) within ovary contains mega spore mother cells which undergoes reduction division to form four haploid megaspores. Three of them disintegrate and only one survive, which after successive mitotic division develop into seven-cell (eight nucleated) structure called female gametophyte or embryo sac. Out of seven three cells found toward chalaza and called **antipodal cells**, three cells found toward micropyle called egg apparatus, two of which termed **synergid cells** and one comparatively large cell is **egg cell** and one central cell with two nuclei called polarnuclei which after fusion called secondary nuclei or definitive nuclei. All the cells except **secondary nuclei** are haploid. This multicellular structure of ovule is termed **embryo sac** which constitutes the female gametophyte of angiosperm, male and female gametophyte dependent on sporophyte.



Double fertilization:

After pollination, pollen grains transfer and settle on the stigma of carpel. The pollen tube grows through the tissue of the stigma and style than enters the ovary with tube nucleus at its tip. When the tip of the pollen tube reaches on ovule, it enters the micropyle and then discharges the two sperm cells into the female gametophyte (embryo sac). One sperm fertilizes the egg cell to form zygote ($2n$) which further develops into an embryo sporophyte. By the time fertilization occurs the two polar nuclei of the female gametophyte have combined to form a diploid fusion nucleus with which the second sperm unites to form a triploid nucleus structure. This nucleus undergoes a series of division and a triploid cell, called endosperm is formed. The endosperms function in the food for the embryo.

Food storage organ provide food to embryo at the time of germination. Double fertilization is a special type of fertilization which occurs only in angiospermic plants. During this process a sperm fuses with the ovum to form oospore. The other sperm fuses with secondary nucleus to form triploid endosperm nucleus.

After fertilization the ovule matures and develop into seed while ovary develops into the fruit. The fruit not only helps to protects the seeds from desiccation during the early development, but often also facilitates their dispersal by various means.

Activity:

Arrange a field trip and collect Cassia, Brassica, Achyranthus, Marcus, Helianthus flowers. Identify different type of inflorescence.

Collect flower of Rose, Cassia fistula, Solanum nigrum and Avena sativa. Describe their flowers

8.4.5 Angiosperms as successful group of land Plants:

The life cycle of angiosperm shows that angiosperms are so well adapted to life on land. Their major advantage over other plants is related to their reproduction. Here they are better adapted in three important ways.

1. The gametophyte generation is very reduced. It is always protected inside sporophyte tissue, on which it is totally dependent. In mosses and liverwort where the gametophyte is conspicuous, while in seedless vascular plants it is a free-living prothallus. The gametophyte is susceptible to drying out.
2. Fertilization is not dependent on water as it is in other plant groups, where sperms swim to the ovum. The male gametes of seed plants are non-motile and are carried within pollen grains that are suited for dispersal by wind, insects, water and animals. Finally transfer of male gametes after pollination takes place by means of pollen tubes. The ova being enclosed within ovules.

3. Conifers and flowering plants produce seeds. Development of seeds is made possible by the retention of ovules and their contents on the parents sporophyte

8.4.6 Inflorescence:

It is mode of branching of floral axis having a group of flower, or it may be defined as arrangement of group of flowers on a floral axis. There are two types of inflorescence racemose and cymose. In racemose inflorescence, main axis called **peduncle** continues to grow. The flower develop in **acropetal succession** and opening of flower is centripetal. Whereas in cymose inflorescence, main axis soon stops growing. The flower develop in basipetal succession and opening of flower is centrifugal.

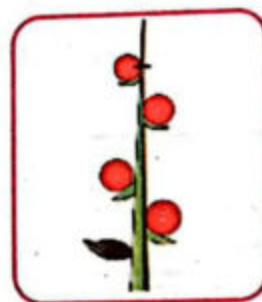
Kinds of racemose inflorescence:

Peduncle elongated:

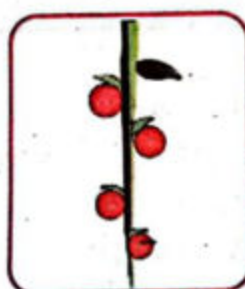
Receme is an inflorescence in which flowers are pedicellate and bisexual e.g., Gold mohar whereas in **spike**, the flowers are sessile and bisexual (amaranthus) **Catkin** is another kind of inflorescence in which flowers are sessile and unisexual (Mulberry) whereas in **spadix**, flowers are covered over by one or many large bracts called **spathes** (e.g Banana)



Raceme



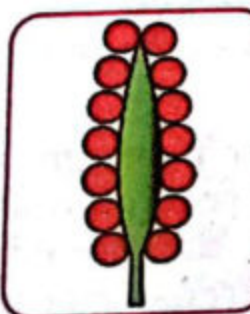
Spike



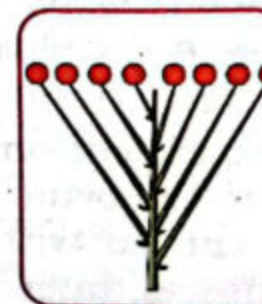
Catkin



Scorpioid cyme



Spadix



Corymb



Fig 8.9 Types of inflorescence



Peduncle shortened:

Corymb is an inflorescence in which flowers have pedicels of unequal length lower flowers having large pedicels and upper ones having small (e.g. Iberis) when pedicels are of same length and arise from common point, the type is known as **umbel** (corianders)

Peduncle flattened:

Head (capitulum) flattened peduncle has a mass of small sessile flowers (florets) with one or more whorls of bracts at the base forming an involucre. The florets are commonly of two kinds namely ray florets (marginal) strap shaped and disc- florets (Central tubular ones) e.g sunflower, zinnia, marigold etc.

Spikelet Inflorescence:

It is kind of racemose inflorescence. There are three bracts at its base called **glumes**. The lower two without flowers are called empty lemma, there is small bracteoles called **Palea**. Flowers, covered by their respective lemma and palea. This type of inflorescence is characteristic feature of family Poaceae.

Kinds of cymose inflorescence:

Uniporous (Monochasial) cyme:

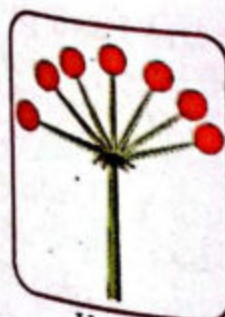
Main axis soon ends into a flower and produces only one lateral branch at a time ending in a flower. The succeeding lateral branches again follow the same mode of producing flower. If the succeeding branches are produced on alternate sides, it is called **scorpioid cyme** (cotton , forget-me-not) . Whereas, if the succeeding branches are produced on same side, it is called **helicoid cyme** (sundew).



Spikelet



Helicoid



Umbel



Biparous



Fig 8.10 Types of cymose inflorescences



Biparous (dichasial) Cyme:

Main axis soon terminate into a flower and produces two flower. This mode is followed by each succeeding flower (Pink night-jasmine).

8.4.7 Significance/benefits of angiosperms for humans:

Angiosperms are very important to human. They have number of uses as food; specially cereals from the grass family. Among most economically important grains are corn (Zeamays), barley (Hordium), oats (Avena), wheat (Triticum) and rice (Aryza).

Vegetables:

There is a variety of vegetables, different in forms and nutritional content, and are grown for one or more of their parts the flowers, shoots, leaves or the underground parts. E.g., Potato, tomato, eggplant and chilli pepper from the potato family (Solanaceae), legumes and beans (Fabaceae), pumpkin, melon and gourd from the squash family (Cucurbitaceae), cabbage, cauliflower, radish and mustard seeds from mustard family (Brassicaceae).

Fruits:

A large number of fruits belongs to rose family (Rosaceae), e.g. Almonds, apples, apricots, cherries, peaches, pears, raspberries and strawberries. Citrus fruits from family Rutaceae, bananas from family Mucaceae and Papaya from family Caricaceae.

Medical/Pharmaceutical plants:

Flowering plants have great pharmaceutical and medical value. Citrus fruits are source of vitamin C, aspirin from bark of willows, narcotics such as opium from opium poppy, quinone from cinchona. Arq-e-gulab extracted from rose (*Rosa indica*) is used as eye drop and for soothing. The pulp of amaltas (*Cassia fistula*) acts as purgative. *Atropa belladonna* of family Solanaceae, *Solanum nigrum* locally called mako used as pain reliever and anti-inflammatory. *Ocimum tenuiflorum* commonly called tulsi used in many herbal medicines to help treat asthma, bronchitis, cold and flu. *Glycyrrhiza glabra* locally called mulhethi traditionally used to treat many diseases particularly in respiratory disorders.

Angiosperm compounds that are highly toxic to humans have proved to be effective in the treatment of certain forms of cancer, such as acute leukemia (vincristine from the Madagascar periwint).



SUMMARY

- Plants are multicellular eukaryotes with photosynthetic nutrition.
- The story of plant revolution is mainly of gradually improving adaption to life on earth.
- Plants show heteromorphic alternation of generation and zygote retained and develops into embryo.
- Bryophytes are small, photosynthetic, free-living gametophytes that lacks vascular tissues.
- Heterogamy is the most successful kind of reproduction that has evolved in bryophytes.
- All the biologists agree that the land plants and animals evolved from aquatic ancestors.
- Land invading plants adapted themselves first to amphibious habits and then developed a complex terrestrial form of life.
- Vascular plants characterized by the presence of vascular tissues i.e. Xylem and Phloem. The dominant generation of living vascular plant is the sporophytes, existing as free-living plants.
- Ferns are the most diverse and conspicuous of the spore-producing vascular plants with about 12000 living species.
- The gametophyte of Pteridophyte (Ferns) is called Prothallus, The Prothallus is independent, autotrophic, heart shaped, dorsoventrally flatland lying prostrate on wet substratum.
- The seed plants have been by for the most successful in fully exploiting the terrestrial environment. They first appeared in the late Devonian, and in the carboniferous they soon replaced the lycopsids and sphenopsids as the dominant land plants, a position they still hold today.
- The seed plants have traditionally divided into two groups. 1. The Gymnospermae 2. The Angiospermae. Gymnospermae have naked seeds because ovules are not covered by ovary while Angiospermae have their seeds enclosed in fruits because ovules are covered by ovary.
- Double fertilization is the specific type of fertilization which occurs only in angiospermic plants. During this process a sperm fuses with the ovum to form ouspore (zygote). The other sperm fuses with secondary nucleus to form triploid endosperm nucleus.
- Flower may be described as compressed reproductive shoot with four whorls of modified (floral) leaves called sepals, petals, stamens and carpels; and which after pollination and fertilization produces seeds within fruits.
- Inflorescence is made of branching of floral axis having a group of flower. The inflorescence may be recemose or cymose.

EXERCISE

1. Encircle the correct choice

- (i) The rapid diversification of plants occurred in the period:

(a) Silurian	(b) Devonian
(c) Carboniferous	(d) Paleozoic
- (ii) Spores of land plants tough, flexible and resistant to biodegradation by:

(a) Lignin	(b) Chitin
(c) Waxes	(d) Sporopollenin
- (iii) Hepaticae, Anthocerotae and Moss are the classes of:

(a) Angiosperms	(b) Gymnosperms
(c) Tracheophytes	(d) Bryophytes
- (iv) The most successful kind of reproduction is:

(a) Heterogamy	(b) Oogamy
(c) Anisogamy	(d) Isogamy
- (v) The dominant generation of living vascular plants existing as free-living plants is:

(a) Sporophyte	(b) Gametophyte
(c) Tracheophyte	(d) Pteridophyte
- (vi) The most primitive living vascular plants is:

(a) Lycophyta	(b) Sphenophyta
(c) Psilophyta	(d) Filicinophyta
- (vii) The gametophyte of Fern is called:

(a) Archegonium	(b) Antheridium
(c) Cones	(d) Prothallus
- (viii) Double fertilization is the characteristic feature of:

(a) Bryophytes	(b) Gymnosperms
(c) Angiosperms	(d) Ferns
- (ix) Resins, turpentine and pine oil are obtained from:

(a) Rose	(b) Onion
(c) Pins	(d) Banana
- (x) In banana tree, flowers are covered over by one or many large brackets called:

(a) Spathes	(b) Spadix
(c) Capitulum	(d) Palea



2. **Write short answers of the following questions:**

1. How life cycle of plants shows alternation of generations?
2. Why bryophytes are called non vascular plants?
3. How plants cope when invaded the land from the sea?
4. Why heterogamy is important for plants?
5. Why flower is called a reproductive part of plant?
6. Why fertilization in angiosperm is called double fertilization?
7. How seeds are evolved?
8. Why gymnosperms have naked seeds but not angiosperms?

3. **Write detailed answers of the following questions:**

1. Define plants. Discuss evolutionary relation in plants.
2. Write down general characteristics of plants.
3. Describe lifecycle of Moss with diagram.
4. Why vascular plants are considered more successful land plants?
5. How leaves are evolved?